

TASKS

Question:1

Open your own laptop or desktop (or use a lab machine) and use CPU-Z to find out the exact RAM type (DDR3, DDR4, or DDR5), frequency (in MHz), and total installed memory. Take a screenshot of the Memory tab and note these details?

Answer: used CPU-Z on my laptop to check the details of the installed RAM. I opened the Memory tab in CPU-Z and noted the following information:

RAM Type: DDR4

RAM Frequency: 3200 MHz

Total Installed Memory: 8 GB

Question:2

Physically examine a RAM module and a storage drive (HDD or SSD) available to you. Identify and write down the form factor (e.g., DIMM, SO-DIMM for RAM; 2.5-inch, M.2 for storage) and the connector type (e.g., SATA, M.2, PCIe) for each?

Answer: I physically examined the RAM module and storage drive available in My laptop.

Component	Form Factor	Connector Type
Ram	SO-DIMM	DDR4 SO-DIMM
Storage drive	M.2 2280 SSD	M.2 PCLE/NVME

RAM: The laptop uses a SO-DIMM RAM module, which is a smaller RAM form factor designed for laptops. It uses a DDR4 SO-DIMM connector.

Storage: The laptop uses an M.2 2280 SSD. It connects through an M.2 PCIe/NVMe interface, which provides fast data transfer.

Question:3

Create a comparison table with at least 4 points contrasting HDD and SSD based on speed, durability, cost, and typical use cases. For each point, give a real-world example (e.g., 'SSD: Used in gaming laptops for fast load times')?

Answer: Point	HDD	SSD
Speed	Slower - used in old PCs For storage.	Faster - used in gaming laptop for quick loading
Durability	Less durable due to moving Parts.	More durable because it has no moving parts.
Cost	Cheaper for large Storage.	More expensive than HDD.
Use	Used for backups, movies And large files.	Used in laptops, gaming PCs and video editing.

Question:4

Calculate the minimum storage capacity needed for each of these scenarios: (a) a YouTuber editing 4K videos weekly, (b) a programmer with lots of code projects and IDEs, (c) a gamer who wants to install 5 AAA games (assume each game is 100GB). Show your calculations?

Answer: (a) YouTuber editing 4K videos weekly:

Assume one 4K video needs 100 GB of storage.

For 4 videos:

$$100 \text{ GB} \times 4 = 400 \text{ GB}$$

Minimum storage = 500 GB

(b) Programmer with many code projects and IDEs:

Assume code, projects, and IDEs need about 200 GB.

Minimum storage = 256 GB

(c) Gamer installing 5 AAA games:

Each game = 100 GB

$5 \times 100 \text{ GB} = 500 \text{ GB}$

Minimum storage = 500 GB

Question:5

Based on your research and calculations, recommend the best storage solution (type and interface) for these three users: (1) a college student using only MS Office and web browsing, (2) a mobile app developer running Android Studio, (3) a competitive gamer who streams on Twitch. Justify your choice for each person?

Answer: 1. College Student - MS Office and Web Browsing Storage: 256 GB or 512 GB SSD Type: SATA SSD Interface: SATA III Reason: It is fast, affordable, and enough for MS Office, web browsing, and documents.

2. Mobile App Developer - Android Studio Storage: 1 TB SSD Type: NV Me SSD Interface: PCIe NV Me Reason: Android Studio, SDKs, emulators, and project files need more storage and fast performance.

3. Competitive Gamer - Twitch Streaming Storage: 1 TB or 2 TB SSD Type: NV Me SSD Interface: PCIe NV Me Reason: Games are large, and fast storage helps with quick loading and smooth gaming/streaming.